Supporting information for "Storing mass-spectrometry data in simple databases enables flexible and intuitive exploration without time or space penalties."

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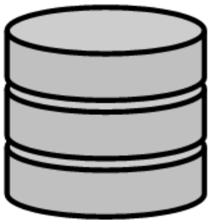

Database
.sqlite
.duckdb

Table: features

feature	name	formula	rt	mz	avg_area	med_snr	med_qscore
Feature	Feature ID	Formula	Retention time	m/z	Average area	Median SNR	Median quality
FT0001	Valine	C5H11NO2	6.5	118.0862	1104948.0	8.79	0.97
FT0002	TMAO	C3H9NO	12.1	76.0760	1729482.0	7.80	0.87
...additional entries...							
FT2311	Unknown	NA	23.7	231.1005	7953192.0	1.72	0.88
FT2312	Unknown	NA	29.5	254.0325	2397.9	3.19	0.24

Table: MS1

filename	scan_num	x_coord	y_coord	mz	int
Source file	Scan number	X coordinate	Y coordinate	m/z ratio	Intensity
Smp_A	1	1	1	60.0452	6618
Smp_A	1	1	1	60.0532	2657
...millions of additional entries...					
Smp_Z	651501	678	327	60.0456	158084
Smp_Z	651501	678	327	60.0531	4673

Table: MS1

filename	scan_num	rt	dt	mz	int
Source file	Scan number	Retention time	Drift time	m/z ratio	Intensity
Smp_A	1	0.10	0.07	60.0452	6618
Smp_A	1	0.10	0.07	60.0532	2657
...billions of additional entries...					
Smp_Z	651501	12.37	6.16	60.0456	158084
Smp_Z	651501	12.37	6.16	60.0531	4673

Table: chrom_peaks

filename	feature	rt_min	rt_max	mz_min	mz_max	peak_area	qscore
Source file	Feature	Min peak RT	Max peak RT	Min peak m/z	Max peak m/z	Area	Quality
Smp_A	FT0001	6.5	7.6	118.0862	118.0865	1104948.0	0.97
Smp_A	FT0002	12.1	13.0	76.0760	76.0762	1729482.0	0.87
...additional entries...							
Smp_Z	FT2311	23.7	25.0	231.1005	231.1006	7953192.0	0.88
Smp_Z	FT2312	29.5	30.6	254.0325	254.0328	2397.9	0.24

Table: rt_cor

filename	scan_num	rt	rt_cor
Source file	Scan number	Retention time	Corrected RT
Smp_A	1	0.10	0.1
Smp_A	1	0.10	0.1
...additional entries...			
Smp_Z	23338	12.37	13.5
Smp_Z	23338	12.37	13.5

Figure S1: Additional possible extensions to the database schema for retention time correction, feature detection and correspondence, ion mobility, and imaging.

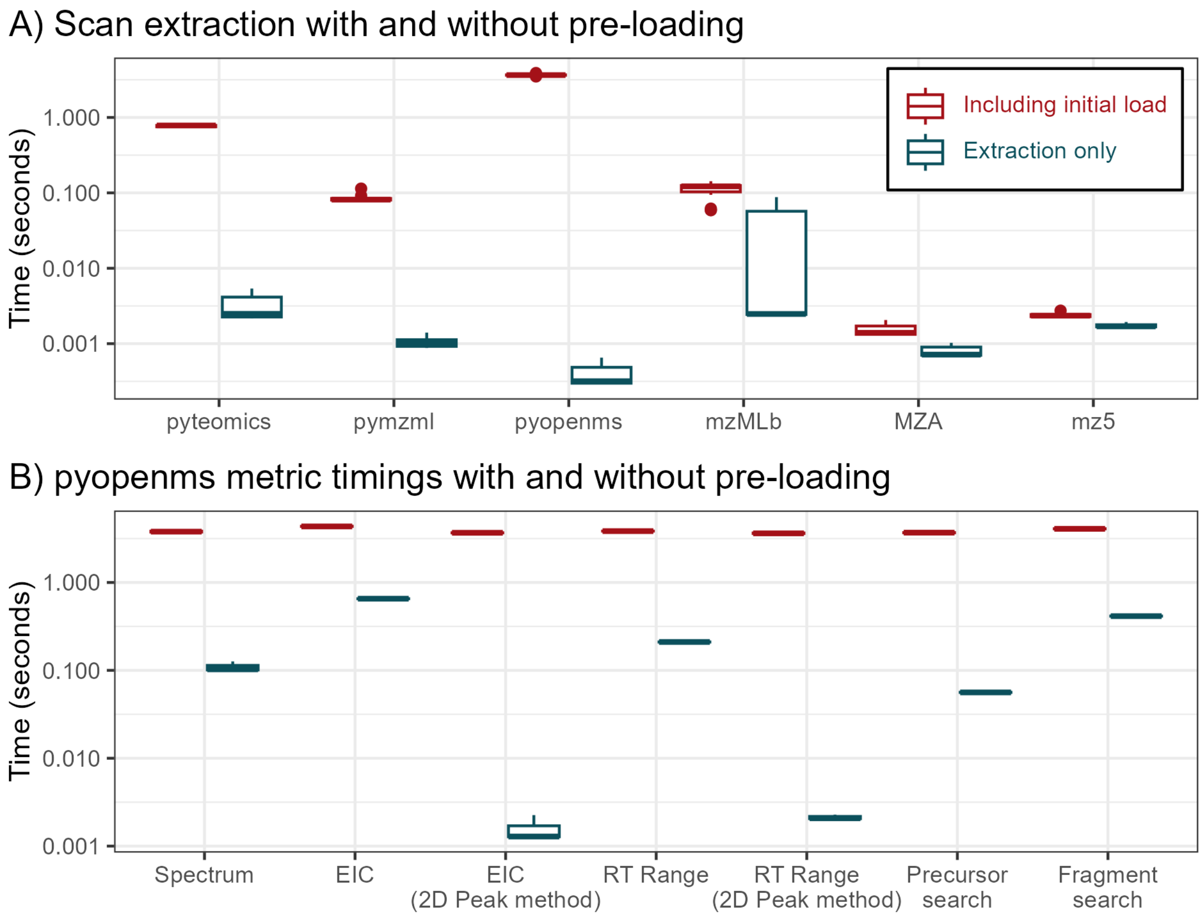

Figure S2: Timing information with and without pre-loading the object into Python prior to extraction. Panel A) shows the number of seconds required to extract a scan for the three mzML parsers plus the mzMLb, MZA, and mz5 file types. Panel B) shows the number of seconds required for pyopenms to perform each of the queries if the initial data load was removed from the timing count. Error bars show 3x replicate queries of 5x different targets.

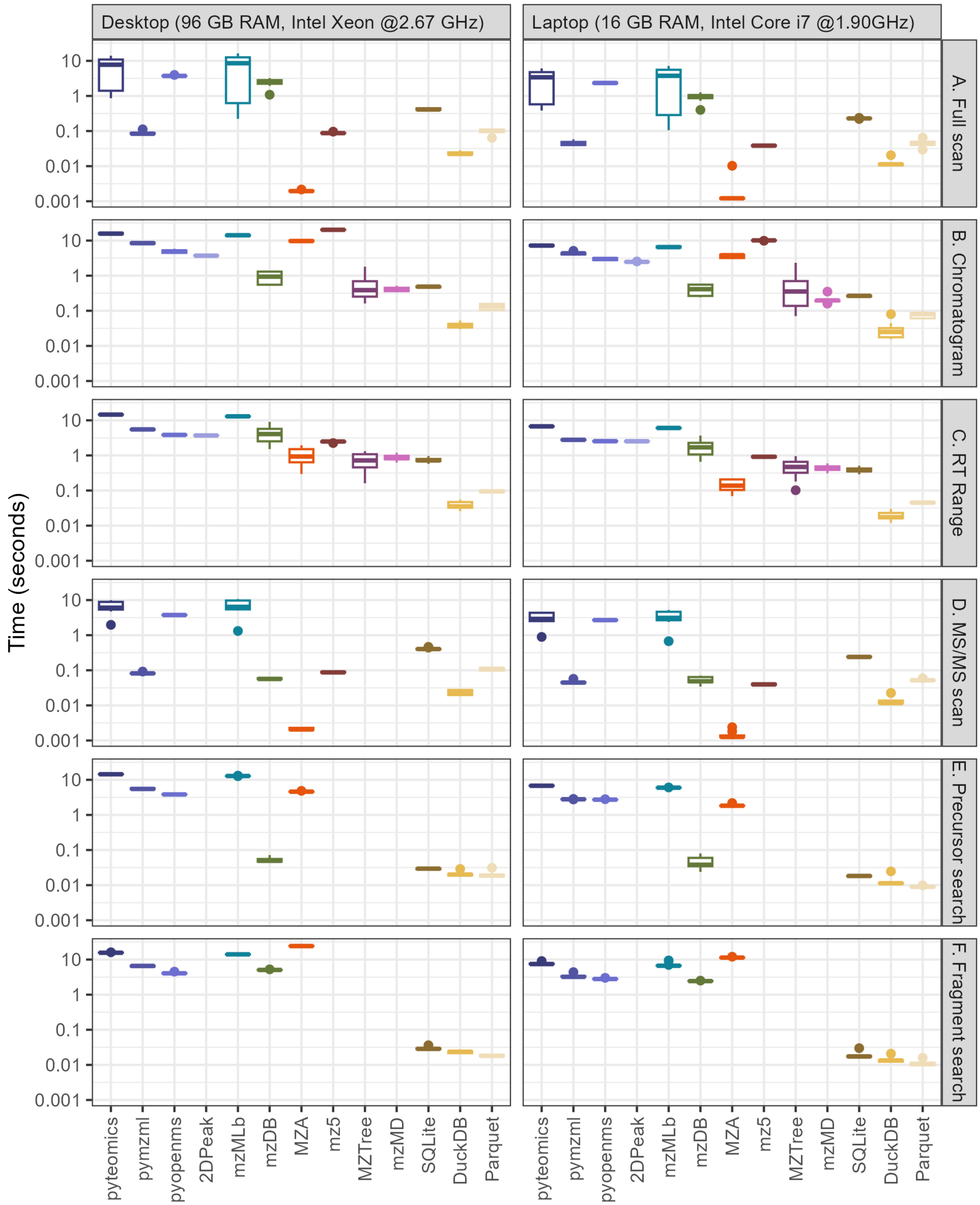

Figure S3: Comparison between an older desktop machine with 96 gigabytes (GB) of RAM and a newer desktop with 16 GB for a single DDA file. Desktop numbers are reproduced from Figure 2 using the same single file as used for the laptop. The panel rows show boxplots representing the time required in seconds to extract a full scan spectrum (A), an ion chromatogram (B), all data within a retention time range (C), an MS/MS scan (D), the fragments of a specified precursor (E), and all precursors with a specified fragment

(F). The error in the boxplot is composed of timing information for 10 repeated queries for a single randomly chosen file, each of a different target scan number, retention time (RT), or m/z, just as in Figure 2.

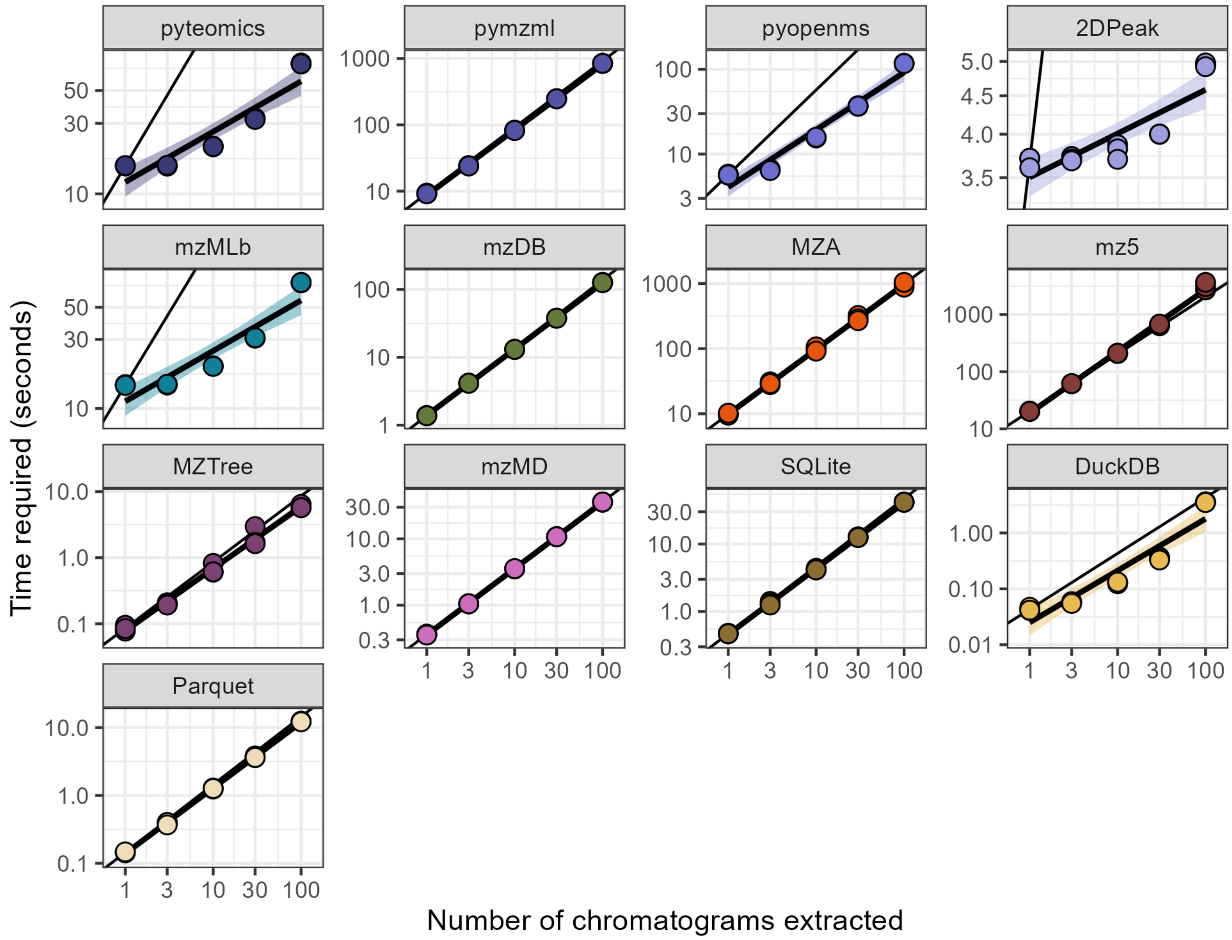

Figure S4: Data from main text Figure 3 with independent y-axes showing scatter plots of the time required to extract multiple chromatograms using various methods on logarithmic axes. Best-fit linear models have been added for each method and are shown behind triplicate timing measurements, while the black line shows a 1:1 slope intercepting the first data point to show nonlinear behavior with additional chromatograms. Transparent intervals around each best-fit line show a single standard error of the mean.

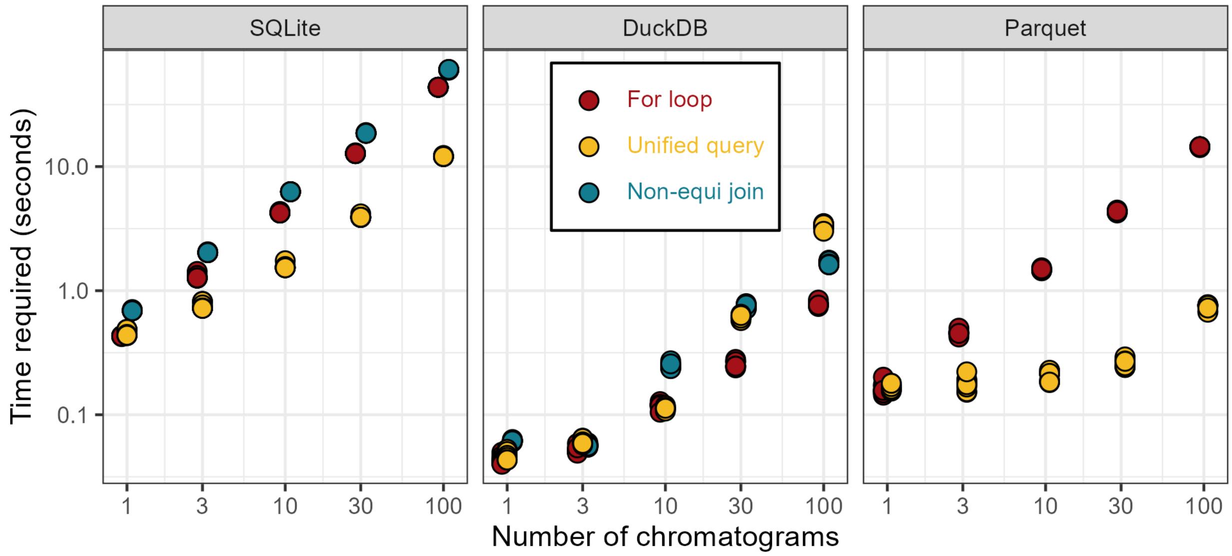

Figure S5: Scatter plot showing time required to extract chromatograms from SQLite, DuckDB, and Parquet databases using different methods denoted by color. Ten replicates are shown for each data point.